

Milestone-2 Report

Objective: The objective of Milestone 2 was to gain practical experience with transformer-based NLP models using the Hugging Face modules. The milestone involved tokenization, embeddings, self-attention, sentence similarity, zero-shot classification, and text generation while comparing these techniques with traditional tfidf based methods.

Dataset: Refer to Milestone-1 report dataset section for dataset information. The same data, provided in the Kaggle competition is used.

Libraries Used:

- datasets: to load the data and preprocess them
- transformers: pretrained tokenizers and models
- sentence-transformers: semantic embeddings
- PyTorch: running transformer models and tensor operations.
- scikit-learn: tfidf implementation and cosine similarity calculation
- pandas: for loading the data

Tasks Performed:

- Used the **datasets** library to load the data and prepare it for tokenization by creating a combined text column consisting of both prompt and the options.
- Tokenized the text using a pretrained tokenizer (**bert-base-uncased tokenizer**) and observed the tokenizer properties.
- Generated contextual embeddings using the bert-base-uncased model and studied its architecture. There were 12 attention heads, and each head has a 64-dimensional vector.
- Observed the attention matrix and saw the float values which represented how much attention a single token gave to all other tokens.
- Used the sentence-embeddings model (**all-MiniLM-L6-v2**) and observed cosine similarity functioning.
- Performed zero-shot classification and used instruction-based generative AI models for text generation from natural language prompts.

Models Used:

- **Tokenization and transformer embeddings:** bert-base-uncased
- **Sentence-embeddings:** all-MiniLM-L6-v2
- **Zero shot classification:** facebook/bart-large-mnli
- **Text generation:** google/flan-t5-small

Key Learnings:

- Understood the difference between traditional TF-IDF based methods and transformer-based embeddings.
- Learned how tokenization converts text into tokens that can be processed by transformer models.
- Understood the purpose of attention heads and how self-attention helps the model understand the relationship between different words in a sentence. Observed and understood the improvement from keyword matching to semantic meaning.
- Understood how in multi-head attention different heads capture different meaning from the sentence.
- Learned how sentence embeddings capture semantic similarity between sentences.

- Understood how zero-shot classification can classify text into new categories without task-specific training.
- Gained practical experience using Hugging Face pipelines for multiple NLP tasks.

Challenges Faced:

- Understanding the attention matrix and transformer outputs was difficult initially because of their complex structure.
- Running different Hugging Face pipelines required handling of model names and dependencies.

Conclusion: Milestone 2 provided practical exposure to transformer-based NLP models and the Hugging Face ecosystem. Through questions involving tokenization, contextual embeddings, self-attention, semantic similarity, zero-shot classification, and text generation, a deeper understanding of modern NLP techniques was developed. This milestone established a strong foundation for using pretrained transformer models in NLP tasks.